

Professional Summary

- Around 4 years of experience as a Machine Learning Engineer specializing in natural language processing (NLP), Artificial Intelligence, Deep Learning, Machine Learning, Data Mining, Advanced analytic, Statistical modeling, GenAI (LLM), and computer vision.
- Proficient in implementing and fine-tuning deep learning algorithms, including Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), LSTM, Transformers, BERT, GPT, LLaMA, and RAG (Retrieval-Augmented Generation).
- Expertise in building various machine learning models using algorithms such as Linear Regression, Logistic Regression, Naive Bayes, Support Vector Machines (SVM), Decision trees, KNN, K-means Clustering, and Ensemble methods (Bagging, Gradient Boosting).
- Strong Knowledge in Python to manipulate data for data loading and extraction and worked with Python libraries like NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, TensorFlow, Ggplot2, OpenCV, PyTorch and NLTK.
- Proficient in visualization tools like Tableau, Power BI, and Excel. Skilled in utilizing cloud platforms, including Azure DevOps and Google Cloud (Vertex AI, Google Cloud Storage) for data engineering and machine learning model deployment.

Education

Master of Science in Computer Science | Majors in **Machine Learning** | Stevens institute of Technology, Hoboken, NJ, USA

Skills

Languages / IDEs: Python, C#, Java, C++, MATLAB, Jupyter Notebook, Google Colab, VS Code, SSMS, R, Git, Docker

Machine Learning: Linear & Logistic Regression, Decision Trees, Random Forests, NumPy, SVM, A/B Testing, Naive Bayes, Gradient Boosting, Feature Engineering, Model Optimization

Deep Learning: CNN, RNN, LSTM, NLP, Large Language Models (LLMs), LangChain, Hugging Face Transformers (BERT, GPT-3), RAG (Retrieval-Augmented Generation), LLaMA, Autoencoders, GANs

Cloud / Visualizations: AWS (EC2, SQS, SNS, CodeDeploy, CloudWatch, API Gateway, S3, SageMaker), GCP (Vertex AI, Google Cloud Storage, BigQuery), Tableau, Power BI, Streamlit

Statistical Techniques: Hypothesis Testing, Data Visualization, Data Modeling, A/B Testing, Model Evaluation, Time-Series Forecasting, Correlation & Regression Analysis, Feature Selection

Packages and Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, NLTK, XGBoost, PyTorch, OpenCV, MLflow, FastAPI

Database and Tools: SQL Server, MySQL, PostgreSQL, MongoDB, Redis, Neo4j, Elasticsearch, Snowflake

Work Experience

JPMorgan Chase & Co., MA

Dec 2024 - Present

Machine Learning Engineer

- Built and optimized machine learning models for financial analytics, achieving a 35% improvement in prediction accuracy and enabling data-driven decision-making.
- Engineered LSTM-based fraud detection models with a 98% accuracy rate, successfully identifying high-risk transactions and enhancing compliance measures.
- Utilized TensorFlow and PyTorch frameworks to design, train, and deploy deep learning models for risk scoring and transaction forecasting, ensuring scalability, reproducibility, and alignment with enterprise MLOps standards.
- Developed and fine-tuned LLMs using GPT and LLaMA architectures for financial document summarization and customer communication insights, processing over 50,000 regulatory filings and client records to enhance contextual understanding of complex financial data.
- Implemented Retrieval-Augmented Generation (RAG) pipelines by integrating internal knowledge bases with large language models, enabling real-time access to 200K+ indexed documents and improving traceability of AI-generated financial insights across multiple business units.
- Established Convolutional Neural Networks (CNNs) for anomaly detection in financial transaction data, achieving a 38% reduction in fraudulent activities and improving overall transaction monitoring efficiency.
- Leveraged cloud-based ML platforms such as AWS SageMaker and Google Cloud AI Platform to accelerate model development and deployment, reducing time-to-market by 30%.
- Partnered with analytics and engineering teams to design ML pipelines and streamline deployment for various financial products.
- Built scalable, high-performance backends using FastAPI and WebSockets for machine learning inference pipelines, leveraging multiprocessing for a 2x speed improvement and real-time monitoring.

Zensar Technologies, India

Jan 2020 - Jul 2023

AI & Deep Learning Developer

- Designed and fine-tuned ML models, like BERT and RNN-based NER, achieving high accuracy for intent classification and key entity extraction.
- Developed automated pipelines using OCR, RAG, and OpenCV, enhancing text extraction accuracy by 98% and reducing data retrieval latency.
- Boosted ML model efficiency by 38% through algorithm optimization, quantization, and pruning while improving anomaly detection and image recognition accuracy.
- Built and deployed models using XGBoost, CNN, and time-series analysis to optimize business operations, increasing sales demands.
- Deployed scalable models on AWS, leveraging CloudWatch for system monitoring and reducing issue resolution time by 30%.
- Utilized Pandas, NumPy, SciPy, Scikit-learn, PyTorch, and OpenCV to streamline image preprocessing, feature extraction, and deep learning model development.
- Implemented MLOps pipelines using Docker, MLflow, and Git to automate model versioning, deployment, and performance tracking, supporting over 25 production models and ensuring consistent delivery across cloud and on-premise environments.
- Applied Explainable AI (XAI) techniques such as SHAP and LIME to interpret model predictions, providing transparency in high-stakes decision-making systems and increasing stakeholder trust in AI-driven solutions.
- Utilized unsupervised approaches such as k-means, Gaussian Mixture Models (GMM), and Encoder-Decoder architecture, resulting in a 20% increase in the precision of text embeddings for evaluation.
- Collaborated on developing a customer service chatbot using Large Language Models (LLMs), integrating with Confluence to retrieve and parse knowledge base articles, enhancing customer satisfaction and service efficiency by 25%.
- Conducted a comprehensive EDA on social media posts, extracting text, metadata, and NLP features. Developed a Long Short-Term Memory (LSTM) model for multi-class sentiment prediction, achieving an 89% validation F1-score.